Activity2 Relationship Workshop

Q1. Scatter Plot

R code:

```
set.seed(1234553)

n <- 24

sales <- round(rnorm(n, 85.0, 18.0), 1)

z <- (sales - 85.0) / 18.0

footfall <- round(300 + 200.0 * (-0.75 * z + 0.00 * z^2 + 0.3) + rnorm(n), 0)

footfall[13] <- round(footfall[13] * 3.2 + 200.0, 0)

df <- data.frame(sales, footfall)

library(ggplot2)

model1 <- lm(footfall ~ sales, data = df)

model2 <- lm(footfall ~ sales + I(sales^2), data = df)

ggplot(df, aes(x = sales, y = footfall)) +

  geom_point(size = 3, color = "blue") +

  geom_smooth(method = "lm", formula = y ~ x, se = FALSE, color = "red") +

  geom_smooth(method = "lm", formula = y ~ x + I(x^2),

              se = FALSE, color = "darkgreen") +

  labs(title = "Weekly Sales vs Footfall",

       x = "Sales (thousand units)",

       y = "Footfall (visitors per day)") +

  theme_minimal()

cat("Correlation:", round(cor(df$sales, df$footfall), 2), "\n")

cat("Linear slope:", round(coef(model1)[2], 2), "\n")

cat("Linear R-squared:", round(summary(model1)$r.squared, 3), "\n")
```

```
cat("Quadratic R-squared:", round(summary(model2)$r.squared, 3), "\n")

outlier <- which.max(abs(rstandard(model1)))

cat("Outlier row:", outlier, "\n")

print(df[outlier, ])

if (summary(model2)$r.squared > summary(model1)$r.squared) {

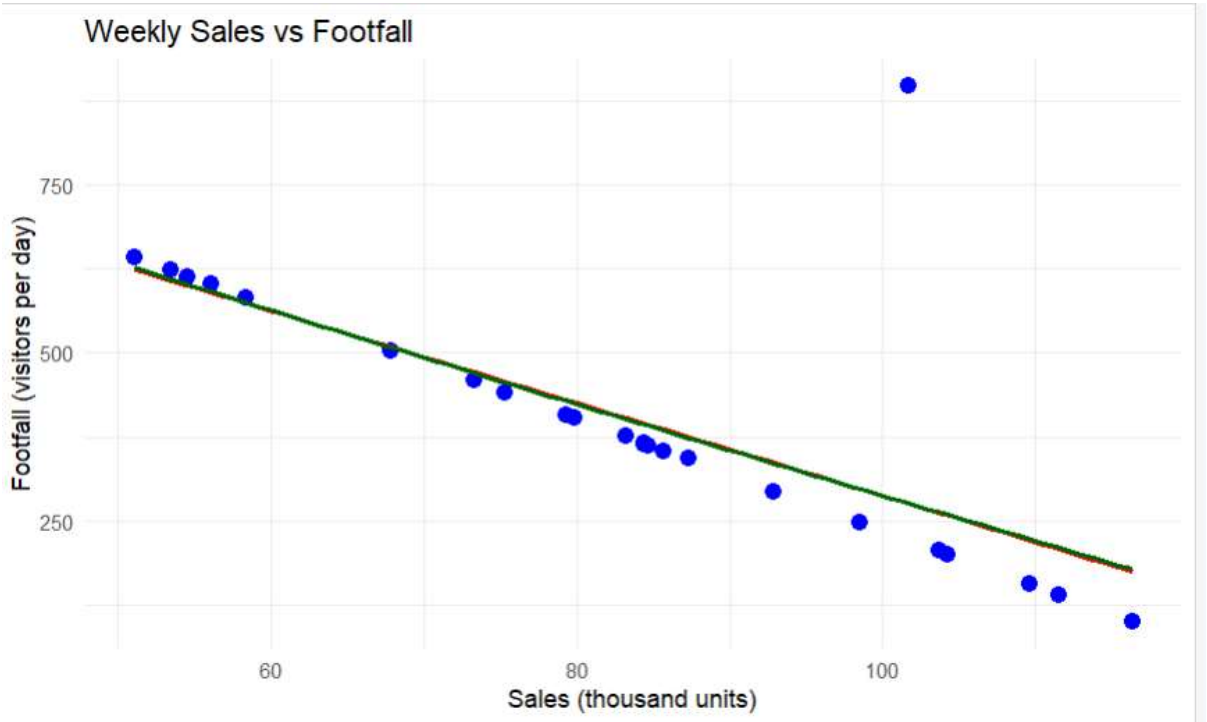
  cat("Quadratic model fits better.\n")

} else {

  cat("Linear model fits better.\n")

}
```

Output:



Q2. Correlogram

R code:

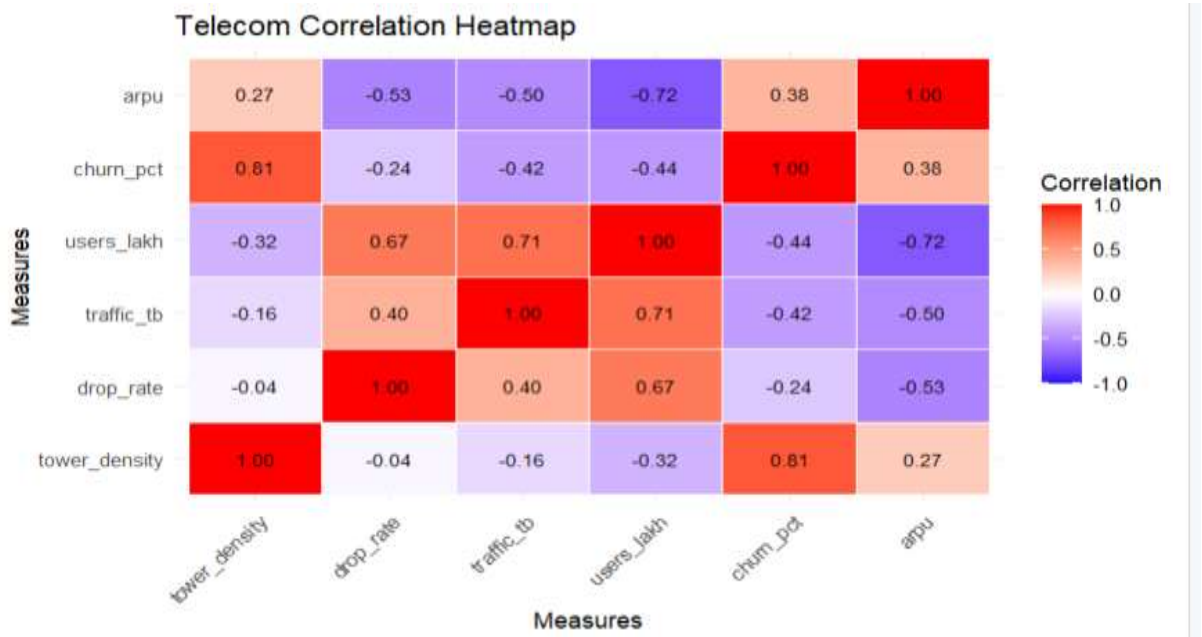
```
set.seed(1235125)

n <- 30
```

```
f1 <- rnorm(n)
f2 <- rnorm(n)
tower_density <- round(14.0 + 4.0 * (0.90 * f2 + 0.44 * rnorm(n)), 2)
drop_rate <- round(2.6 + 0.7 * (0.75 * f1 + rnorm(n)), 1)
traffic_tb <- round(320 + 70.0 * (0.76 * f1 + 0.65 * rnorm(n)), 0)
users_lakh <- round(100 + 24.0 * f1 + rnorm(n), 1)
churn_pct <- round(1.8 + 8.5 * (0.88 * f2 + 0.60 * rnorm(n)), 1)
arpu <- round(185 + 25.0 * (-0.70 * f1 + 0.65 * rnorm(n)), 0)
df <- data.frame(tower_density, drop_rate, traffic_tb,
                 users_lakh, churn_pct, arpu)
library(ggplot2)
library(reshape2)
cor_matrix <- round(cor(df), 2)
print(cor_matrix)
cor_data <- melt(cor_matrix)
ggplot(cor_data, aes(Var1, Var2, fill = value)) +
  geom_tile(color = "white") +
  geom_text(aes(label = sprintf("%.2f", value)), size = 3) +
  scale_fill_gradient2(low = "blue", mid = "white",
                       high = "red", midpoint = 0,
                       limits = c(-1, 1)) +
  labs(title = "Telecom Correlation Heatmap",
       x = "Measures", y = "Measures",
       fill = "Correlation") +
```

```
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
cor_matrix[lower.tri(cor_matrix, diag = TRUE)] <- NA
pairs <- which(abs(cor_matrix) == max(abs(cor_matrix), na.rm = TRUE),
               arr.ind = TRUE)
print(cor_matrix)
cat("Strongest correlation:", cor_matrix[pairs[1, 1], pairs[1, 2]], "\n")
weak <- which(abs(cor_matrix) < 0.2 & !is.na(cor_matrix), arr.ind = TRUE)
if (nrow(weak) > 0) {
  print(data.frame(
    Variable1 = rownames(cor_matrix)[weak[, 1]],
    Variable2 = colnames(cor_matrix)[weak[, 2]],
    Correlation = cor_matrix[weak]
  ))
}
```

Output:



Q3. Dimension Reduction

R code:

```
set.seed(1225182)

n <- 36

grp <- rep(c("Group A", "Group C", "Group B"), each = 12)

g <- rep(c(-1, 0, 1), each = 12)

footfall_k <- round(120 + 30 * rnorm(n) + 1.2 * g, 1)

spend_rs <- round(3200 + 600 * rnorm(n) + 1.2 * g, 0)

stay_nights <- round(2.8 + 0.7 * rnorm(n) + 0.8 * g, 1)

rating <- round(7.6 + 0.8 * rnorm(n) + 0.8 * g, 1)

occupancy_pct <- round(68 + 10 * rnorm(n) + 1.2 * g, 1)

complaints <- round(4.0 + 1.2 * rnorm(n) + 0.8 * g, 1)

df <- data.frame(footfall_k, spend_rs, stay_nights,

                 rating, occupancy_pct, complaints)

pca <- prcomp(df, scale. = TRUE)

variance <- pca$sdev^2 / sum(pca$sdev^2)

print(round(variance * 100, 2))

cat("PC1 variance:", round(variance[1] * 100, 2), "%\n")

cat("PC2 variance:", round(variance[2] * 100, 2), "%\n")

cat("Components needed for 80% variance:",

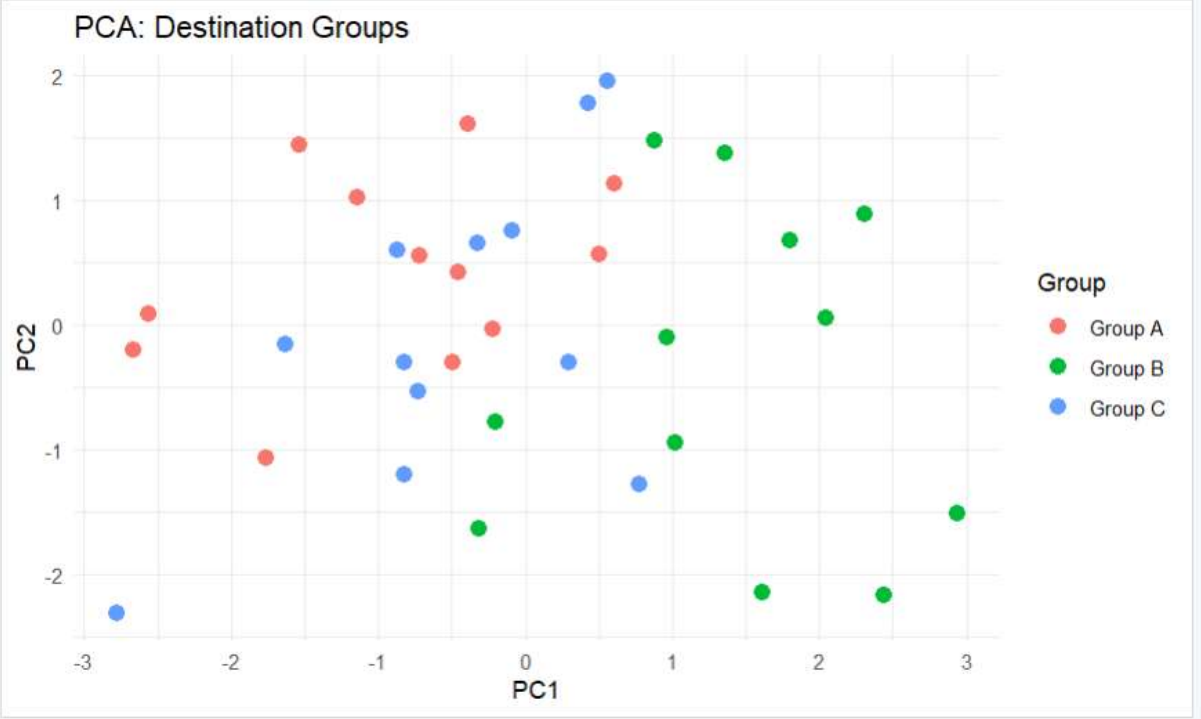
    which(cumsum(variance) >= 0.80)[1], "\n")

library(ggplot2)

scores <- data.frame(pca$x[, 1:2], grp = grp)
```

```
ggplot(scores, aes(PC1, PC2, color = grp)) +
  geom_point(size = 3) +
  labs(title = "PCA: Destination Groups",
       x = "PC1", y = "PC2", color = "Group") +
  theme_minimal()
cat("PC1 loadings:\n")
print(round(pca$rotation[, 1], 3))
cat("PC1 and PC2 R-squared:\n")
print(round(sum(variance[1:2]), 3))
install.packages("Rtsne")
library(Rtsne)
tsne <- Rtsne(scale(df), dims = 2, perplexity = 5,
              check_duplicates = FALSE, pca = FALSE)
tsne_data <- data.frame(tsne$Y, grp = grp)
ggplot(tsne_data, aes(X1, X2, color = grp)) +
  geom_point(size = 3) +
  labs(title = "t-SNE: Destination Groups",
       x = "Dimension 1", y = "Dimension 2",
       color = "Group") +
  theme_minimal()
```

Output:



Q4. Network Graph

R code:

```
set.seed(12251212)

nodes <- c("Mumbai", "Delhi", "Ahmedabad", "Hyderabad",

           "Bengaluru", "Kolkata", "Chennai")

pairs <- t(combn(nodes, 2))

edges <- data.frame(

  from = pairs[, 1],

  to = pairs[, 2],

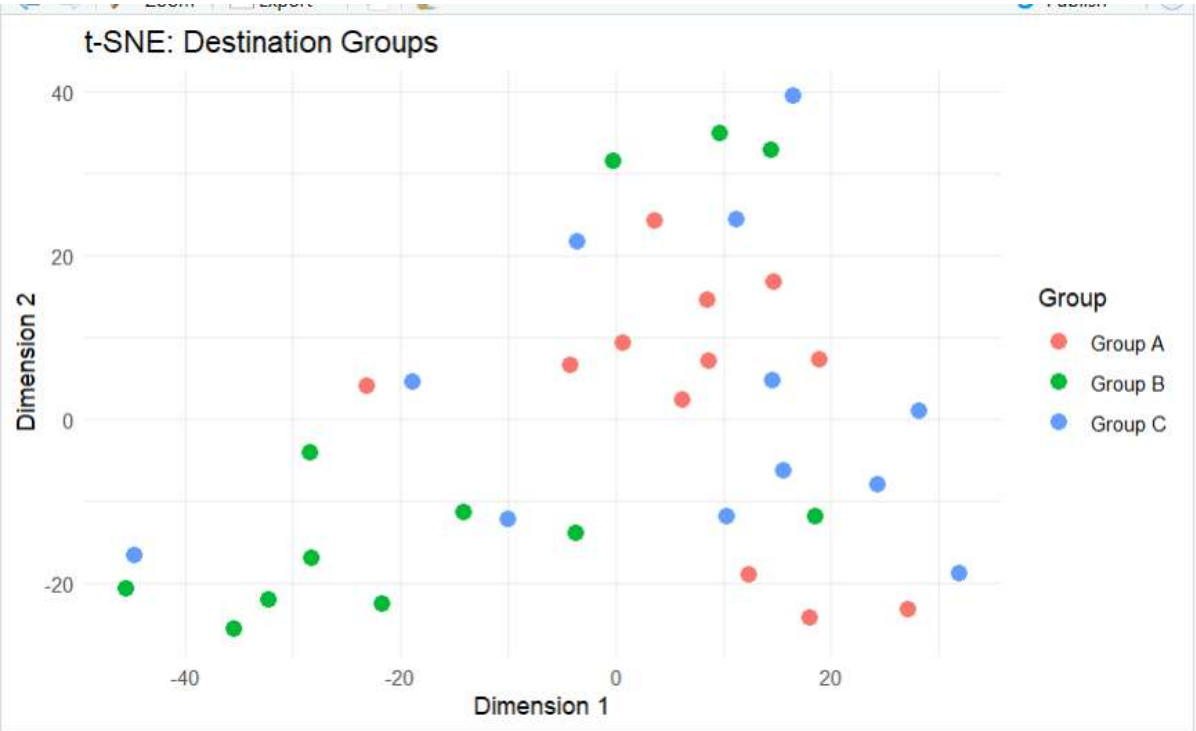
  weight = sample(10:100, nrow(pairs), replace = TRUE)

)

library(igraph)
```

```
g <- graph_from_data_frame(edges, directed = FALSE,
                           vertices = nodes)
plot(g,
     vertex.color = "lightblue",
     vertex.size = 25,
     vertex.label.cex = 0.8,
     edge.width = E(g)$weight / 20,
     edge.label = E(g)$weight,
     main = "Weighted Parcel Delivery Network"
```

Output:



Q5. Paired/Slope chart

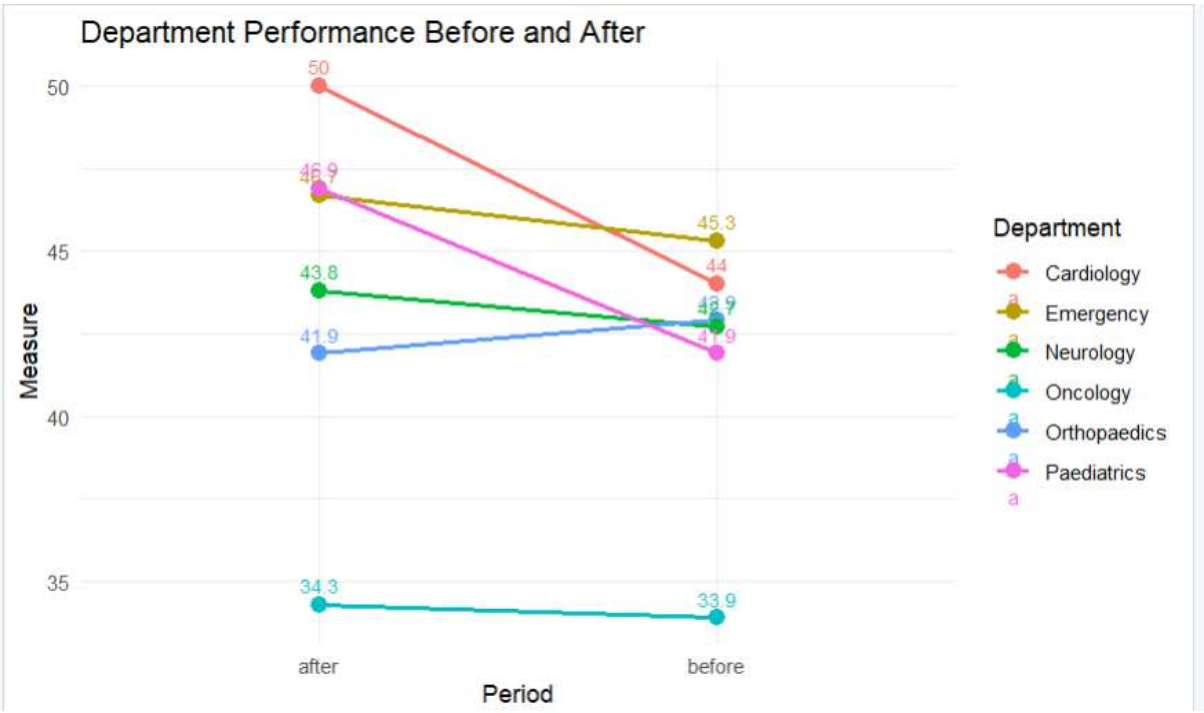
R code:

```
entity <- c("Emergency", "Orthopaedics", "Neurology",
            "Paediatrics", "Oncology", "Cardiology")
```

```
before <- round(rnorm(6, 42.0, 5.0), 1)
change <- round(c(1.4, -1.0, 1.1, 5.0, 0.4, 6.0), 1)
after <- before + change
df <- data.frame(entity, before, after)
print(df)
library(ggplot2)
library(tidyr)
long <- pivot_longer(df, cols = c(before, after),
                     names_to = "period", values_to = "value")
ggplot(long, aes(x = period, y = value, group = entity,
                 color = entity)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  geom_text(aes(label = value), vjust = -0.8, size = 3) +
  labs(title = "Department Performance Before and After",
       x = "Period", y = "Measure", color = "Department") +
  theme_minimal()
cat("Correlation:", round(cor(before, after), 3), "\n")
cat("Mean change:", round(mean(after - before), 2), "\n")
test <- t.test(after, before, paired = TRUE)
print(test)
if (test$p.value < 0.05) {
  cat("The change is statistically significant.\n")
} else {
```

```
  cat("The change is not statistically significant.\n")
}
```

Output:



Q6. Bubble Chart

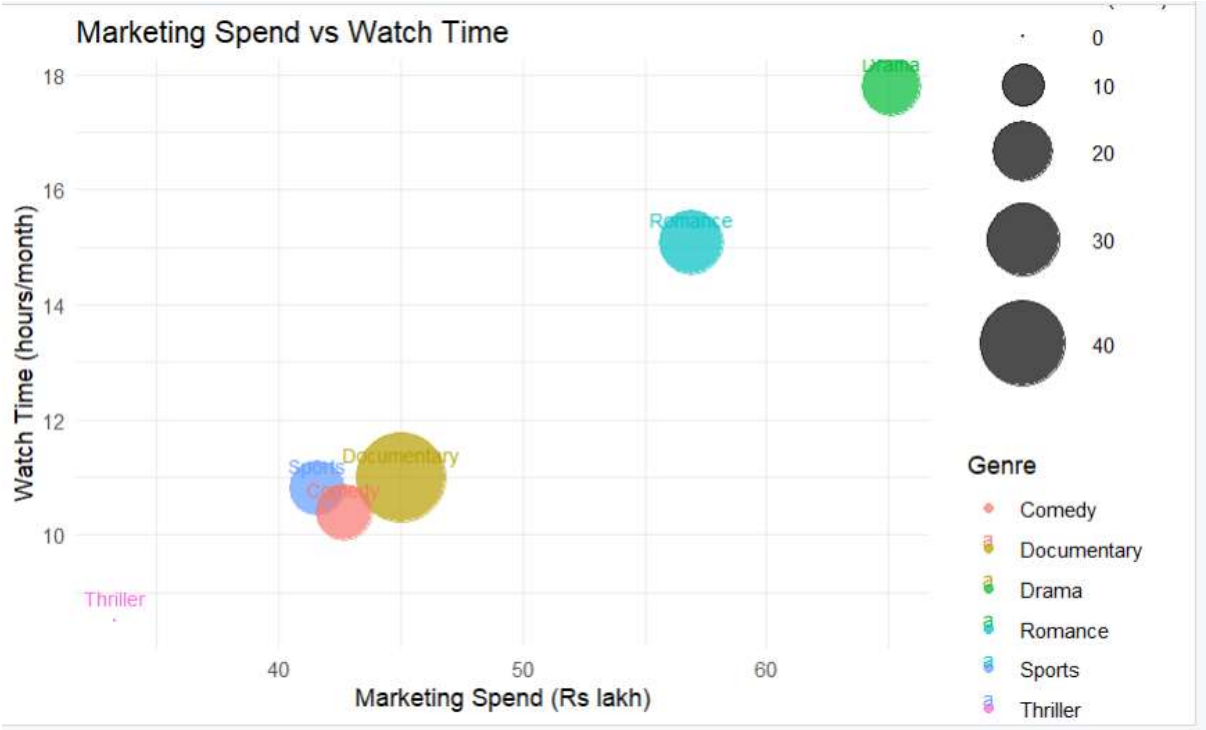
R Code:

```
set.seed(1225318)
entity <- c("Thriller", "Drama", "Documentary",
            "Sports", "Romance", "Comedy")
n <- 6
z <- rnorm(n)
spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 3.8 * z + 0.80 * rnorm(n), 1)
subs_lakh <- round(18.0 + 4.8 * rnorm(n), 1)
subs_lakh[3] <- round(18.0 + 3.8 * rnorm(1), 1)
spend_lakh[3] <- 45.0
```

```
watch_hrs[3] <- 11.0
subs_lakh[3] <- 45.0
df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
print(df)
library(ggplot2)
ggplot(df, aes(x = spend_lakh, y = watch_hrs,
               size = subs_lakh, color = entity)) +
  geom_point(alpha = 0.7) +
  geom_text(aes(label = entity), vjust = -1, size = 3) +
  scale_size_area(max_size = 18, name = "Subscribers (lakh)") +
  labs(title = "Marketing Spend vs Watch Time",
       x = "Marketing Spend (Rs lakh)",
       y = "Watch Time (hours/month)",
       color = "Genre") +
  theme_minimal()
cat("Correlation (spend and watch time):",
    round(cor(spend_lakh, watch_hrs), 3), "\n")
cat("Correlation (spend and subscribers):",
    round(cor(spend_lakh, subs_lakh), 3), "\n")
z_size <- (subs_lakh - mean(subs_lakh)) / sd(subs_lakh)
largest <- which.max(subs_lakh)
cat("Largest bubble:", entity[largest], "\n")
cat("Largest subscriber value:", subs_lakh[largest], "\n")
cat("Z-score of largest bubble:", round(z_size[largest], 2), "\n")
```

```
cat("Times the median size:",
    round(subs_lakh[largest] / median(subs_lakh), 2), "\n")
```

Output:



Q7. Hexbin Plot

R code:

```
set.seed(1225371)
n <- 3000
core <- runif(n) < 0.85
loan_lakh <- round(ifelse(core, rnorm(n, 38.0, 5.4),
                          runif(n, 11.0, 65.0)), 1)
income_k <- round(ifelse(core,
                         52.0 + 0.71 * (loan_lakh - 38.0) / 5.4 +
                         sqrt(1 - 0.51) * 7.2 * rnorm(n),
                         runif(n, 16.0, 88.0)), 1)
```

```
df <- data.frame(loan_lakh, income_k)
library(ggplot2)
p <- ggplot(df, aes(x = loan_lakh, y = income_k)) +
  geom_hex(bins = 25) +
  scale_fill_viridis_c(name = "Record count") +
  labs(title = "Hexbin Plot of Loan Amount and Income",
       x = "Loan Amount (Rs lakh)",
       y = "Applicant Income (Rs thousand/month)") +
  theme_minimal()
print(p)
print(ggplot(df, aes(x = loan_lakh, y = income_k)) +
        geom_point(alpha = 0.3) +
        labs(title = "Scatterplot of Loan Amount and Income",
             x = "Loan Amount (Rs lakh)",
             y = "Applicant Income (Rs thousand/month)") +
        theme_minimal())
xbin <- cut(df$loan_lakh, breaks = 10)
ybin <- cut(df$income_k, breaks = 10)
counts <- table(xbin, ybin)
idx <- which(counts == max(counts), arr.ind = TRUE)[1, ]
cat("Busiest cell count:", counts[idx[1], idx[2]], "\n")
cat("Loan range:",
    levels(xbin)[idx[1]], "\n")
cat("Income range:",
```

```
    levels(ybin)[idx[2]], "\n")
cat("Share of all records:",
    round(100 * counts[idx[1], idx[2]] / nrow(df), 2), "%\n")
cat("Distinct loan-income pairs:", nrow(unique(df)), "\n")
```

Output:

